

3.1 & 3.2 The Derivative Function Worksheet

1. Suppose a particle's position is given by the equation $s(t) = 1 - t^2 + 5t$, where s is measured in meters and t is measured in seconds.
- a) Find the average velocity of the particle during its first 4 seconds of motion. State answer with units.

$$\text{average rate of change} = m_{\text{sec}} = \frac{s(4) - s(0)}{4 - 0} = \frac{5 - 1}{4} = 1 \text{ m/s}$$

- b) Find the instantaneous velocity of the particle after 1 seconds. State answer with units.

$$\begin{aligned} \text{instantaneous rate of change} &= m_{\text{tan}} = \lim_{h \rightarrow 0} \frac{s(t+h) - s(t)}{h} = \lim_{h \rightarrow 0} \frac{[1 - (t+h)^2 + 5(t+h)] - [1 - t^2 + 5t]}{h} \\ &= \lim_{h \rightarrow 0} \frac{1 - t^2 - 2th - h^2 + 5t + 5h - 1 + t^2 - 5t}{h} = \lim_{h \rightarrow 0} \frac{-2th - h^2 + 5h}{h} \\ &= \lim_{h \rightarrow 0} (-2t - h + 5) = -2t + 5 \end{aligned}$$

Therefore, the instantaneous rate of change at $t = 1$ is $-2(1) + 5 = 3 \text{ m/s}$

2. Find $f'(x)$ if $f(x) = \frac{1-x}{2+x}$.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1-(x+h)}{2+(x+h)} - \frac{1-x}{2+x}}{h} = \lim_{h \rightarrow 0} \frac{\frac{1-(x+h)}{2+(x+h)} - \frac{1-x}{2+x}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{1-(x+h)}{2+(x+h)} \cdot \frac{2+x}{2+x} - \frac{1-x}{2+x} \cdot \frac{2+x+h}{2+x+h} \right] \\
 &= \lim_{h \rightarrow 0} \frac{(2-2x-2h+x-x^2-xh) - (2+x+h-2x-x^2-xh)}{h(2+x)(2+x+h)} \\
 &= \lim_{h \rightarrow 0} \frac{-3h}{h(2+x)(2+x+h)} = \lim_{h \rightarrow 0} \frac{-3}{(2+x)(2+x+h)} = \frac{-3}{(2+x)(2+x)} = \frac{-3}{(2+x)^2}
 \end{aligned}$$

3. Find an equation that will tell you slope of any tangent line on the graph of $f(x) = 4x^2 - 9$.

This means that we are looking for the derivative.

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[4(x+h)^2 - 9] - [4x^2 - 9]}{h} = \lim_{h \rightarrow 0} \frac{4x^2 + 8xh + 4h^2 - 9 - 4x^2 + 9}{h} \\
 &= \lim_{h \rightarrow 0} \frac{8xh + 4h^2}{h} = \lim_{h \rightarrow 0} (8x + 4h) = 8x
 \end{aligned}$$

4. Use the information found in problem #3 to construct the equation of a line that is tangent to the function $f(x) = 4x^2 - 9$ at the point where $x = \frac{1}{4}$.

To construct the tangent line we need the point (x, y) on f where the tangent line is being constructed.

$$\text{Point: } x = \frac{1}{4} \quad y = f\left(\frac{1}{4}\right) = 4\left(\frac{1}{4}\right)^2 - 9 = -8.75$$

$$\text{Slope: } f'\left(\frac{1}{4}\right) = 8\left(\frac{1}{4}\right) = 2$$

$$\text{Tangent Line: } y - (-8.75) = 2\left(x - \frac{1}{4}\right) \rightarrow y = 2x - 9.25$$

5. Find $f'(x)$ if $f(x) = x^3 - 5x$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[(x+h)^3 - 5(x+h)] - [x^3 - 5x]}{h}$$

You will have to multiple out $(x+h)^3 = (x+h)(x+h)(x+h)$ unless you can utilize the *Binomial Theorem*.

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3 + 5x}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 - 5h + h^3}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh - 5 + h^2) = 3x^2 - 5 \end{aligned}$$

6. Find the derivative of $g(x) = \frac{1}{x+1}$.

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)+1} - \frac{1}{x+1}}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{1}{(x+h)+1} \cdot \frac{x+1}{x+1} - \frac{1}{x+1} \cdot \frac{x+h+1}{x+h+1} \right] \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{x+1 - x - h - 1}{h(x+h+1)(x+1)} \right] = \lim_{h \rightarrow 0} \frac{-h}{h(x+h+1)(x+1)} = \lim_{h \rightarrow 0} \frac{-1}{(x+h+1)(x+1)} = \frac{-1}{(x+1)(x+1)} = \frac{-1}{(x+1)^2} \end{aligned}$$

7. Suppose that we know that $g(x)$ is an *even* function that is differentiable at $x = 1$. we know that $g(x)$ has the following values:

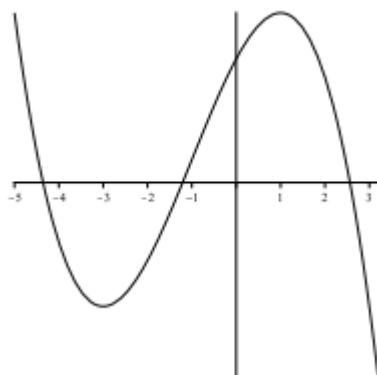
$$g(-4) = 1 \quad g(-2) = 1 \quad g(-1) = 3 \quad g(0) = 3 \quad g(2) = 1 \quad g(3) = 1$$

Find $\lim_{x \rightarrow 1} g(x)$.

Since $g(x)$ is even then $g(x) = g(-x)$. Since $g(x)$ is differentiable at $x = 1$ then $g(x)$ is continuous at $x = 1$.

So, $\lim_{x \rightarrow 1} g(x) = g(1) = g(-1) = 3$.

8. The graph of the function $f(x)$ is shown below.



One of the six graphs below is the graph of $f'(x)$. Circle the graph of $f'(x)$.

